

Python Assignment: Beginner-Level Programming Task

Objective

This assignment will test your understanding of basic Python concepts, including variables, conditionals, loops, functions, and file handling.

Instructions

1. The assignment is individual work.
2. Code comments are mandatory to explain your logic.
3. Submit your Python file with the name Assignment_<YourName>.py.
4. Your program should handle invalid inputs gracefully (use error handling where necessary).
5. Submission deadline: **28-11-2024**.
6. This assignment covers 10% of overall coursework.

Part 1: Basic Programming

Task 1: Arithmetic Operations

Write a Python program that:

1. Takes two numbers as input from the user.
2. Performs addition, subtraction, multiplication, and division.
3. Displays the results in a formatted way.

Task 2: Temperature Converter (10 Marks)

Write a program that converts temperatures between Celsius and Fahrenheit. The program should:

1. Ask the user to choose the conversion type:
 - 1 for Celsius to Fahrenheit
 - 2 for Fahrenheit to Celsius
2. Take the temperature as input and display the converted value.

Task 3: Simple Interest Calculator

Write a program that calculates simple interest based on the formula:

$$S=P \times R \times T$$

Where:

- P is the principal amount.
- R is the rate of interest.
- T is the time (in years).

The program should:

1. Take P, R, and T as inputs.
2. Calculate and display the simple interest.

Part 2: Intermediate Problem Solving

Task 4: Palindrome Checker

Write a program that checks whether a given string is a palindrome. A palindrome is a string that reads the same backward as forward (e.g., "madam", "racecar").

Task 5: Number Guessing Game

Write a Python program for a number guessing game:

1. The program randomly selects a number between 1 and 100.
2. The user must guess the number.
3. After each guess, the program tells the user if their guess is too high, too low, or correct.
4. The game continues until the user guesses the correct number.

Part 3: File Handling

Task 6: Save and Read a File

1. Write a program that asks the user to input 5 lines of text.
2. Save the text to a file named user_text.txt.
3. Read and display the contents of the file.